Date: 2026-05-27

notes on papers

snc

- t1

(2) annotations: Ambridge & Blything 2024.pdf

paper: @

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comment

6

investigating the ability of LLMs to simulate human grammatical acceptability judgments

6

preference for (negative values) periphrastic over transitive causatives

Q: unclear: why is certain, that the model does not simply judge over the grammaticality of the sentences by probabilistic measures? where here is the relation to (ratings), such can we not say the model prefers x% vs. human y% of Z

(1) annotations: Bronk, Zwitserlood, Bölte (2013).pdf

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4

frequency classes can be obtained for each word relative to the frequency of the masculine definite article “der,” which is the most frequent word in German

(4) annotations: Frankowsky_2022.pdf

paper: Frankowsky (2022)

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2

Frankowsky

#cite_Frankowsky_2022.pdf__

11

the date of attestation for each formation was identified.

17

Data concerning the date of attestation were not available for all 2,337 cases of ProtICCs. The year of the utterance could be identified in 772 cases (33.0%).

25

Schäfer, Roland & Bildhauer, Felix. 2012. Building Large Corpora from the Web Using a New Efficient Tool Chain. LREC: 486–493.

(2) annotations: Kodner et al 2023.pdf

paper: @

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2

Under this hypothesis, children generalize from their limited input in specific ways, navigating a constrained space of possible natural language grammars. Consequently, they do not consider all logically possible generalizations that are consistent with their linguistic experience

2

From these two claims, Piantadosi concludes that LLMs show that unconstrained learning from small data is possible

(18) annotations: Piantadosi (2024).pdf

paper: Piantadosi et al. (2024)

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The internal representations of words in these models are stored in a vector space, and the locations of these words include not just some aspects of meaning, but properties that determine how words can occur in sequence (e.g. syntax).

6

they are trained only on text prediction.3This means that the models form probabilistic expectations about the next word in a text and they use the true next word as an error signal to update their latent parameters

#kybernetik

6

how training a neural network on text prediction could lead it to discover key pieces of the underlying linguistic system.

#q: still doubts if just being able to reproduce lx system (grammar) yet shows an understanding of the logic (beyond probabilities) of the underlying (system). #gpt q: write a story in english based on the german / japanese / russian grammar. see: https://esteeschwarz.github.io/SPUND-LX/pages/004/lxtech003.html

7

Linguistic corpora are a low-dimensional projection of both syntax and thought, so it is not implausible that a smart learning system could recover at least some aspects of these cognitive systems from watching text alone

8

language models should be treated as bona fide linguistic theories

10

a learning mechanism configures the model itself in the space of theories in order to satisfy the desired objective function.

11

rephrasing of his (and others') comments on language learning – essentially that one should not study an unconstrained system because it will not explain why languages have the particular form that they do.9

12

In particular, we may view each model or set of modeling assumptions as a possible hypothesis about how the mind may work. Testing how well a model matches human-like behavior then provides a scientific test of that model's assumptions. This is how, for example, the field has discovered that attentional mechanisms are important for performing well

12

helping to delineate what is possible: Could it be possible to discover hierarchy without it being built in? Could word prediction provide enough of a learning signal to acquire most of grammar? Could a computational architecture achieve competence on WHquestions without movement, or use pronouns without innate binding principles?

13

Surprisingly or not, the language model vectors appear to encode at least some aspects of semantics

16

Part of making these models work well was in determining how to encode semantic properties into vectors, and in fact initializing word vectors via encodings of distribution semantics

18

Probability is central for models because a probabilistic prediction essentially provides an error signal that can be used to adjust parameters that themselves encode structure and generating processes

18

they are based in a continuous calculus that allows multiple influences to have a gradient effect on upcoming linguistic items.

18

The fact that predictions are probabilistic is useful because it means that the underlying representations are continuous and gradient

#discrete

19

Continuity is important because it permits the models to use gradient methods – essentially a trick of calculus – to compute what direction to change all the parameters in order to decrease error the fastest.

19

prevalence of “relaxation” methods in numerical computing, where an optimization problem with hard constraints is often best solved via a nearby soft, continuous optimization problem

19

The success of gradient models over deterministic rules suggests that quite a lot of language is based on gradient computation

22

aux-inversion

(4) annotations: Smolka et al 2013.pdf

paper: @

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2

full activation of the base when a regularly inflected form is encountered

3

The psycholinguistic implementation of this hypothesis takes the form of socalled “dual-mechanism” models, assuming two innately distinct systems, each incorporating a specific processing style: The default system parses regular forms into their constituent morphemes, whereas the memory system stores and retrieves all exceptions to the default as undecomposed whole words

3

certain ERP components are associated with certain linguistic processes, so that the components may be used as indicators for particular processes.

3

dual-mechanism model of Ullman and colleagues assumes that regular forms are parsed in frontal areas, whereas irregular forms are retrieved from declarative memory in parietal areas

(1) annotations: Wilkinson et al. 2016.pdf

paper: @

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1

Findability, Accessibility, Interoperability, and Reusability

(21) annotations: Zinsmeister 2013.pdf

paper: Zinsmeister (2013)

page

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comment

1

Heike Zinsmeister

#cite_Zinsmeister 2013.pdf__

2

Our corpus-based study is grounded in the approach of Distributional Semantics, which assumes that the meaning of a word can be modeled by collecting information about its distribution in a corpus

2

we distinguish between three different semantic compound types. The first type is traditionally referred to as a determinative (or endocentric) compound (cf. Olsen 2000). Its meaning is compositionally derived from the meanings of its parts: The meaning of the head noun denotes the type of referent, and the meaning of the modifier noun modifies this in some semantic relation to the head noun, such that it further specifies or modifies the meaning of the head noun.

2

The distributional information is collected by examination of co-occurring words, structures, or relations. The sum of this contextual information can be viewed as a kind of mold, a container that shapes the meaning of the word in a characteristic fashion.

3

The final semantic type is traditionally called the copulative (or dvandva) compound

3

includes all noun-noun compounds that are lexicalized such that their meanings are opaque with respect to the meanings of their parts.(..)(..)They are semantically

3

is the least frequent type of compound in German. In Fanselow's (1981b) definition, the meaning of a copulative compound is created by applying the basic relation 'AND' to the meanings of its parts

3

The second semantic type is traditionally referred to as the possessive (also bahuvrihi or exocentric) compound.

4

the distributions of the target nouns are vectors.

5

In such a vector space, each context word corresponds to a dimension, and the distribution of a target word is a vector in this multidimensional space. The similarity between two words can then be measured in terms of the distance between their vectors in the space

5

The similarity of two distributions is interpreted as the distance between their vectors. A common way to measure this distance is to calculate the cosine of the angle between the two vectors (e.g., Erk 2012: 637). If two vectors point in the same direction, meaning that the angle between them is 0°, then the cosine will be 1. Otherwise, the cosine will be less than 1; it will be - 1 if the two vectors point in exactly opposite directions

5

The basic elements in the standard model are the co-occurrence frequencies of context words. Context is defined as a window on the token string.

6

The Kullback-Leibler divergence

6

skew divergence as a variant of the KullbackLeibler divergence. Instead of calculating the divergence using the observed probability distributions, this function smoothes the values of q, meaning that q will be positive for all dimensions for which p has a positive value

6

Lee (1999, 2001) has introduced skew divergence

7

extracted all words that were annotated as common nouns in the corpus and analyzed them further using SMOR (Schmid et al. 2004

7

skewed distribution of head family size

what?

8

skew divergence distributions D. These scores range from a minimum of 0.093, observed for the divergence of a head from its compound (cf. leftmost boxplot D(c(..)(..)h)), to a maximum of 2.293, observed for the divergence of a compound from its modifier

8

The similarity of head and compound will exceed that of modifier and compound:

8

The similarity of head and compound will not exceed the similarity of modifier and compound:

8

the compound will have a high similarity to both head and modifier:

Frankowsky, Maximilian. 2022. *Extravagant Expressions Denoting Quite Normal Entities: Identical Constituent Compounds in German*. John Benjamins Publishing Company. https://www.degruyterbrill.com/document/doi/10.1075/slcs.223.07fra/html.

Piantadosi, Steven T., Stefan Müller, Moshe Poliak, Geoffrey K. Pullum, Robert D. Van Valin Jr, Dafydd Gibbon, Iris Berent, et al. 2024. “Modern Language Models Refute Chomsky’s Approach to Language.” In *From Fieldwork to Linguistic Theory*, 353–415. Language Science Press. https://langsci-press.org/catalog/book/434.

Zinsmeister, Heike. 2013. “Corpus–Based Modeling of the Semantic Transparency of Noun–Noun Compounds.” In *A Festschrift for Susan Olsen*, edited by Holden Härtl, 303–21. Berlin: Akademie Verlag. https://doi.org/doi:10.1524/9783050063799.303.

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